

Introduction

An important consequence of human-induced climate change is the increase in extreme weather events (EE), such as heat waves, droughts, and heavy rainfall [1]. In this study, we first investigate the occurrence of these events in Brazil between 1940 and 2020, using data from the ERA5 reanalysis database and climate indices recommended by the World Meteorological Organization (WMO). Our analysis reveals increasing trends in temperature extremes, particularly in maximum temperature events (TX90p and TX10p), in accordance to present literature [2]. We then compare these statistical indicators with anomalies detected using unsupervised machine learning, specifically LSTM (Long-Short Term Memory) autoencoders, to further investigate climate anomalies.

Methodology

ERA5 Reanalysis Data

- Fifth generation ECMWF reanalysis database for global climate and weather
- Combination of model data with observations, compiled together with physical laws
- Spatial resolution: 0.25°
- Uncertainty: 0.5°
- Variables: daily temperature and precipitation (1940–2020)

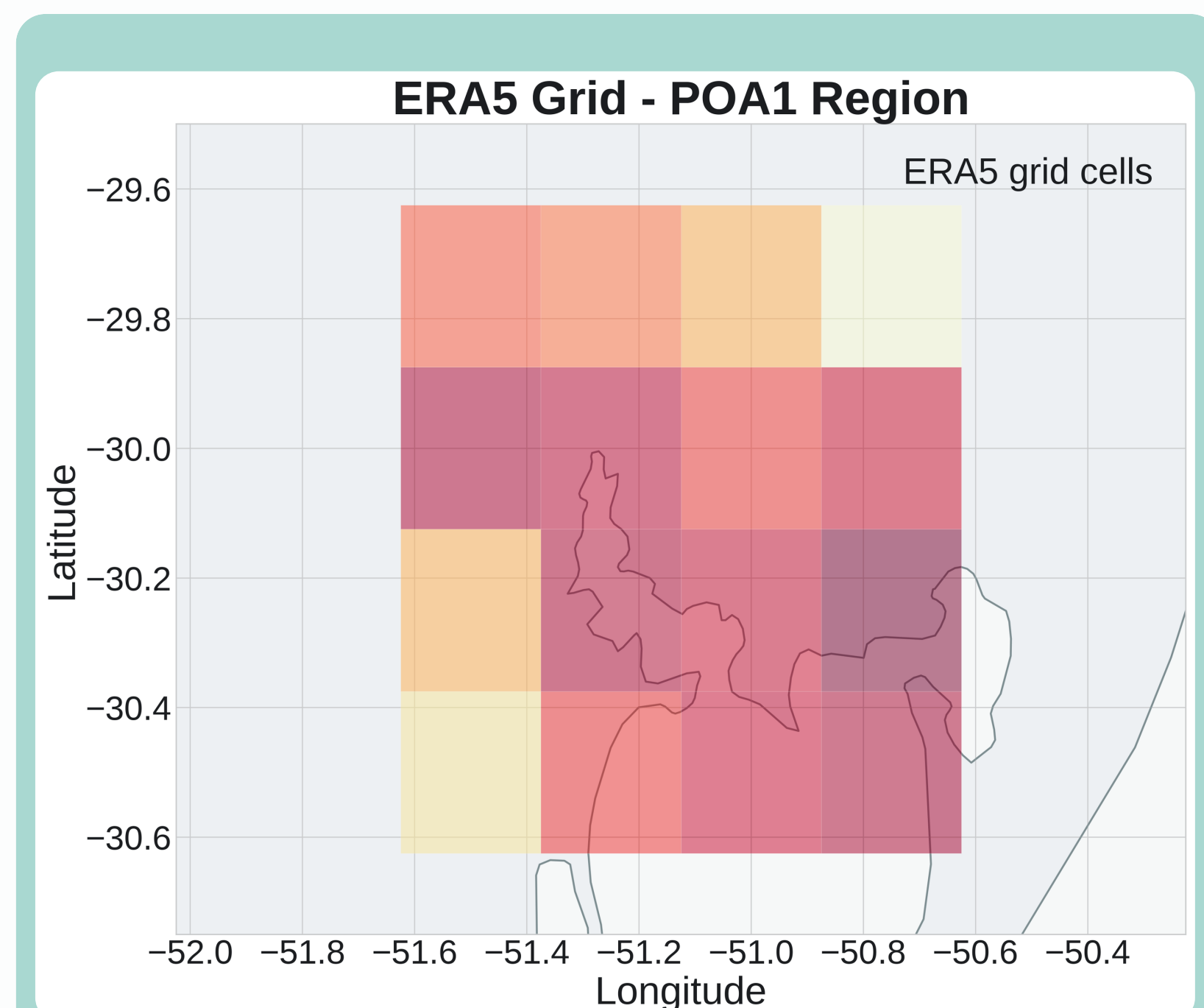


Figure 1: ERA5 grid points in the initial region of analysis.

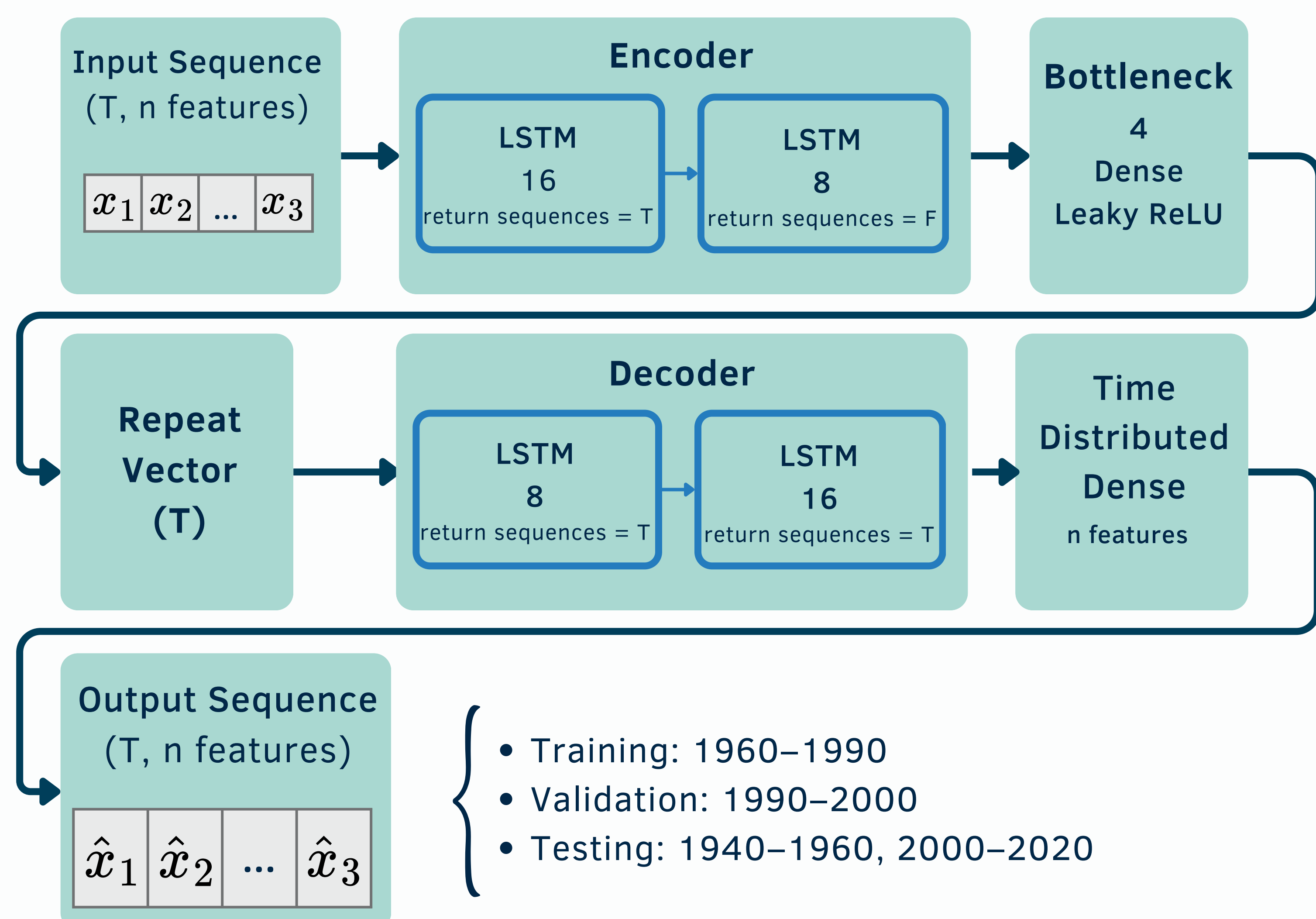
Extreme Event Identification:

- Extreme cold: temperature values below the 10th percentile
- Extreme heat: temperature values above the 90th percentile
- One extreme day = one daily exceedance (TX10p, TX90p)

- Reference period: 1960–1990
- Analysis periods: 1940–1960 and 1990–2020
- Tmax analyzed jointly for 10th and 90th percentiles

LSTM Autoencoders for Anomaly Detection

- Input features: latitude, longitude, time, mean/max/min temperature, precipitation, and dry days
- Model reconstructs normal behavior in climate time series [3]
- High reconstruction error indicates anomaly



Anomaly Detection

- Daily values for features are recovered by mapping each sequence timestep back to its original day and aggregating overlaps.

$$MSE_i = \frac{1}{T} \sum_{t=1}^T (x_{t,i} - \hat{x}_{t,i})^2$$

$$Anomaly_i = \begin{cases} 1, & MSE_i > \tau_i \\ 0, & \text{otherwise} \end{cases}$$

T : sequence length
 $x_{t,i}$: true value of feature i at time step t
 $\hat{x}_{t,i}$: reconstructed value of feature i at time step t
 MSE_i : reconstruction error for feature i
 τ_i : anomaly threshold for feature i (90th percentile) of MSE values

EE and Anomaly Indices

- To compare statistical extreme events and LSTM anomalies, we compute the annual mean frequency of detected events across all ERA5 grid points.
- The same aggregation formula is applied to both EE counts and LSTM anomaly counts.

Mean EE over several ERA5 points:

$$\overline{N_y^{EE}} = \frac{\sum_s N_{s,y}^{EE}}{f_y} \quad \frac{\text{EE count at grid point } s \text{ in year } y}{\text{total number of analyzed grid points in year } y}$$

Results

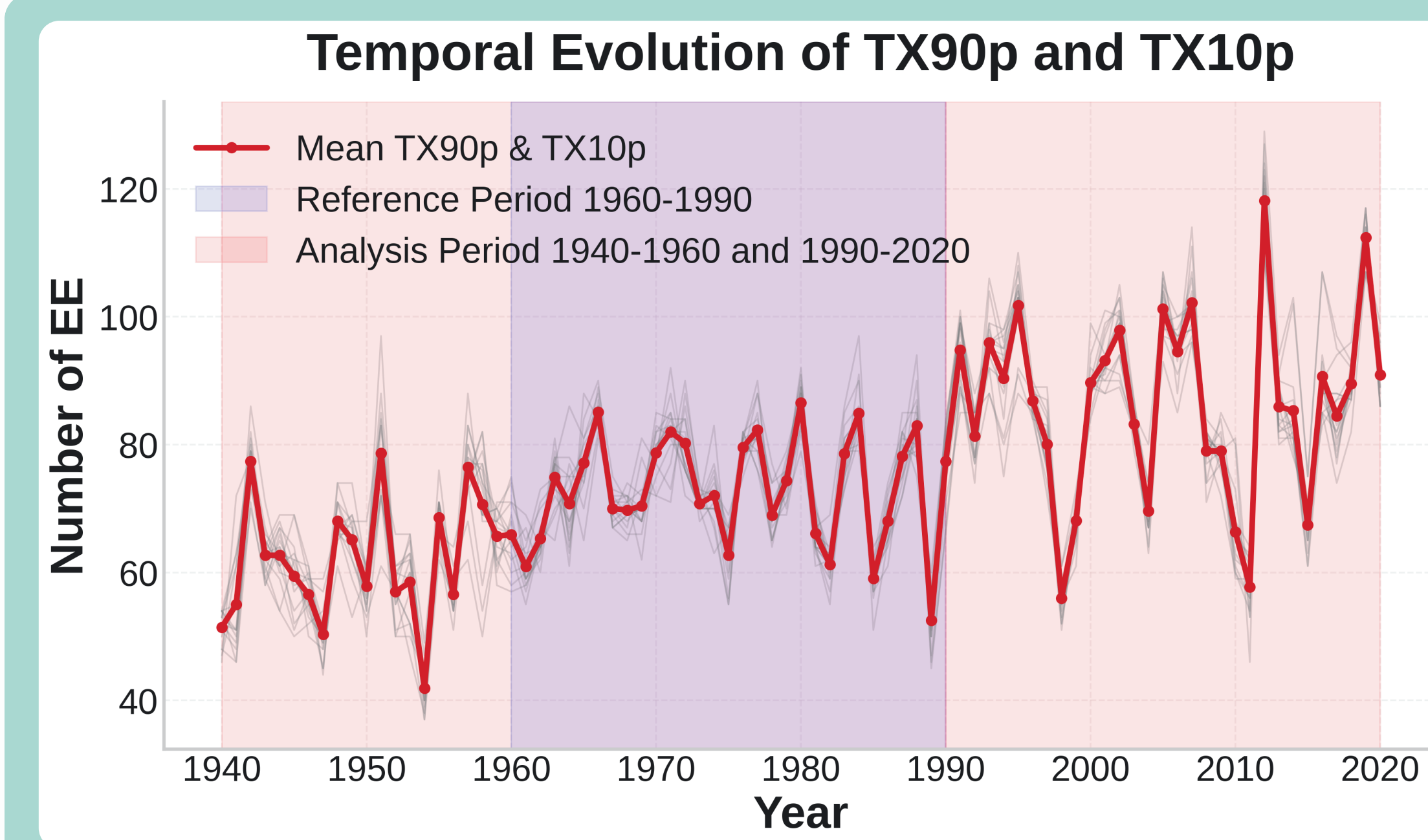


Figure 2:

Temporal evolution of the mean of maximum temperature events (TX90p + TX10p) in the POA1 region. Grey lines indicate the individual ERA5 point data, and the red line is the mean over all points. Periods of analysis are indicated in color blocks. Overall, a clear increase in EE can be noted.

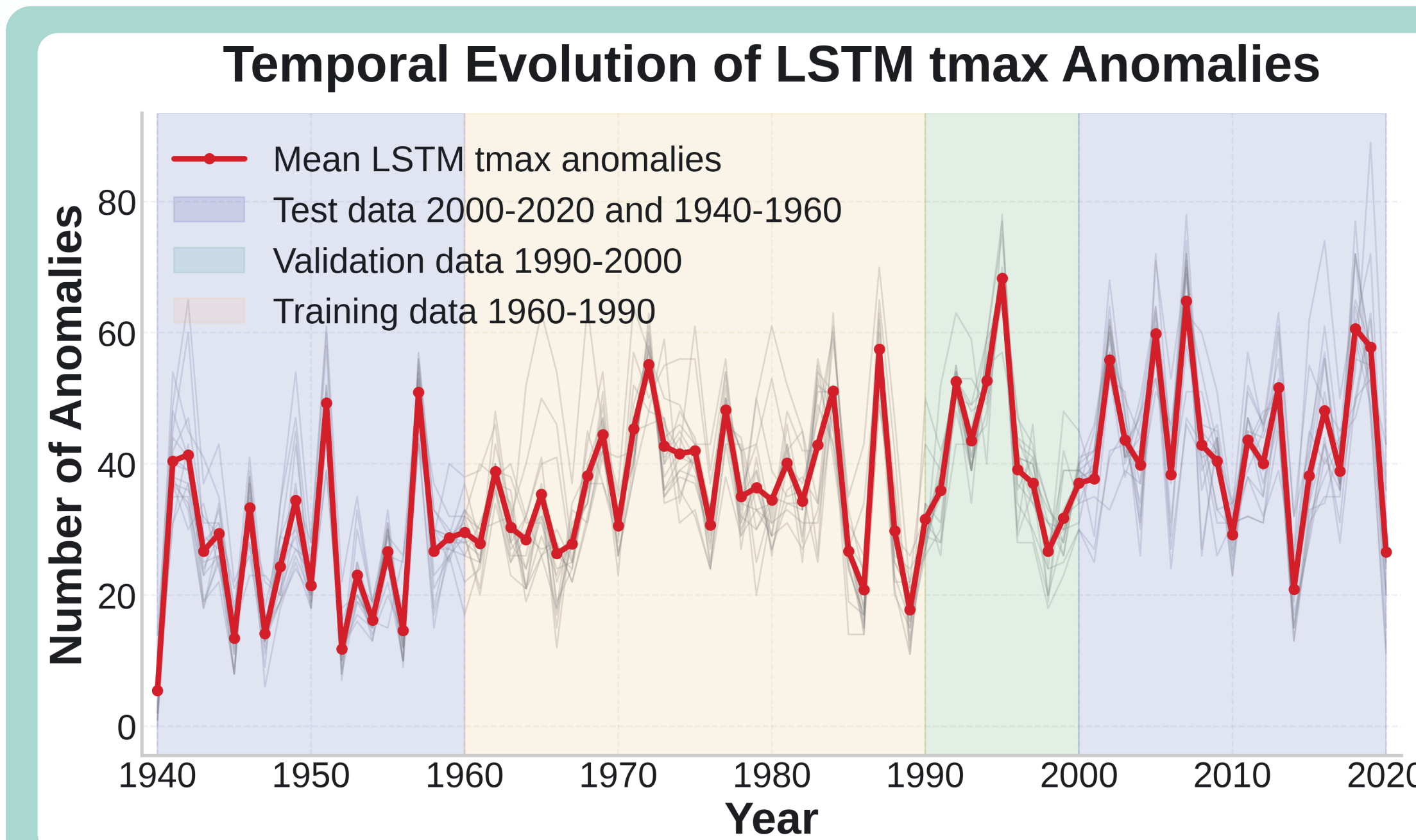


Figure 3:

Temporal evolution of the maximum temperature (tmax) LSTM anomalies detected in the POA1 region. Grey lines indicate the individual ERA5 point data, and the red line is the mean over all points. Training, testing and validation data are indicated in blocks. Overall, a slight increase in anomalies can be noted.

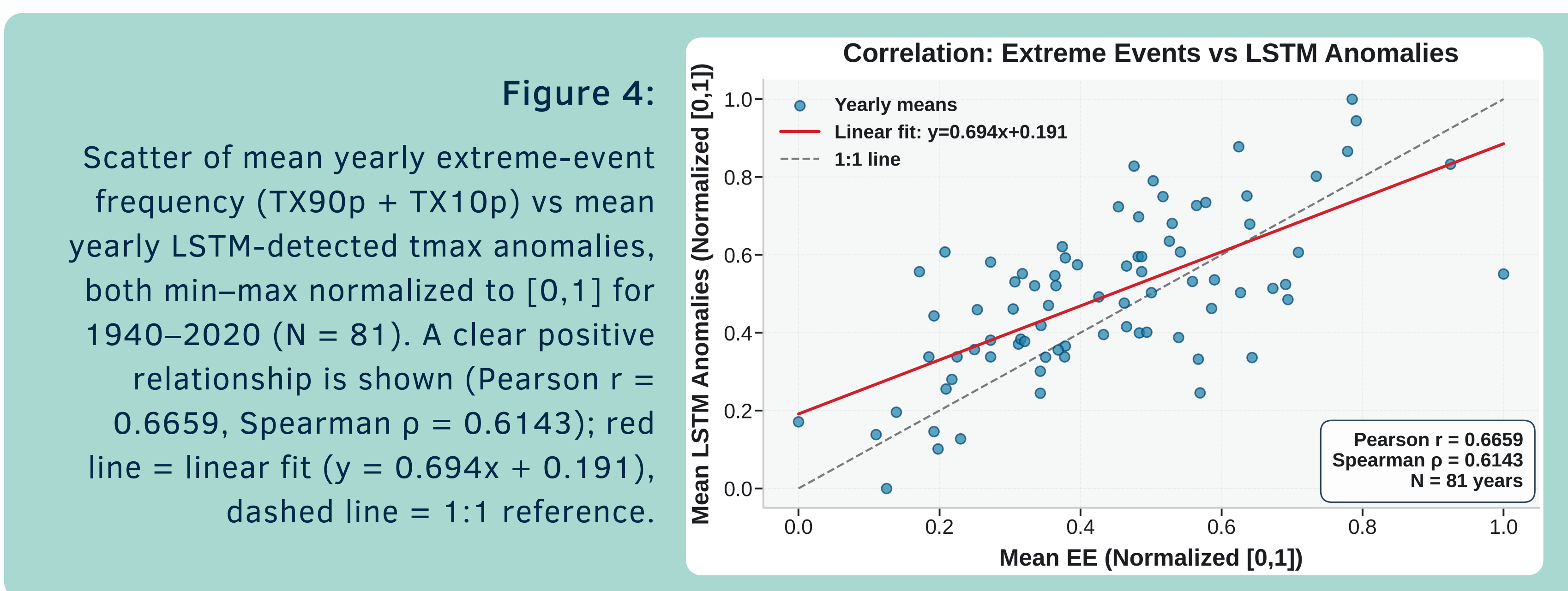


Figure 4:

Scatter of mean yearly extreme-event frequency (TX90p + TX10p) vs mean yearly LSTM-detected tmax anomalies, both min–max normalized to [0,1] for 1940–2020 (N = 81). A clear positive relationship is shown (Pearson $r = 0.6659$, Spearman $\rho = 0.6143$); red line = linear fit ($y = 0.694x + 0.191$), dashed line = 1:1 reference.

Conclusions

Our results show a consistent increase in maximum temperature extreme events, including TX90p and TX10p, as well as a rise in LSTM-detected temperature anomalies over time. Although statistical extreme indices and machine learning anomalies do not describe identical meteorological phenomena, Figure 4 shows a strong relationship between the two approaches. Similar analyses are being extended to other variables, including precipitation and minimum temperature. These findings suggest that AI-based methods can serve as complementary tools for automated extreme-event detection and for improving our understanding of relationships among meteorological variables.

References

- [1] V. Masson-Delmotte, P. Zhai, et al., Climate change 2021: the physical science basis, Cambridge University Press (2021).
- [2] D. Lazzari, A. Garcez, N. Poltozi, G. Pozzi, C. Brito, Climatic Extremes in Brazil: a parallel analysis of Historical Trends and Socioeconomical Impacts, arXiv:2409.16309 (2026)
- [3] Rakesh Shrestha, Mohammadreza Mohammadi, et al., Anomaly detection based on LSTM and autoencoders using federated learning in smart electric grid, Journal of Parallel and Distributed Computing, 2024.